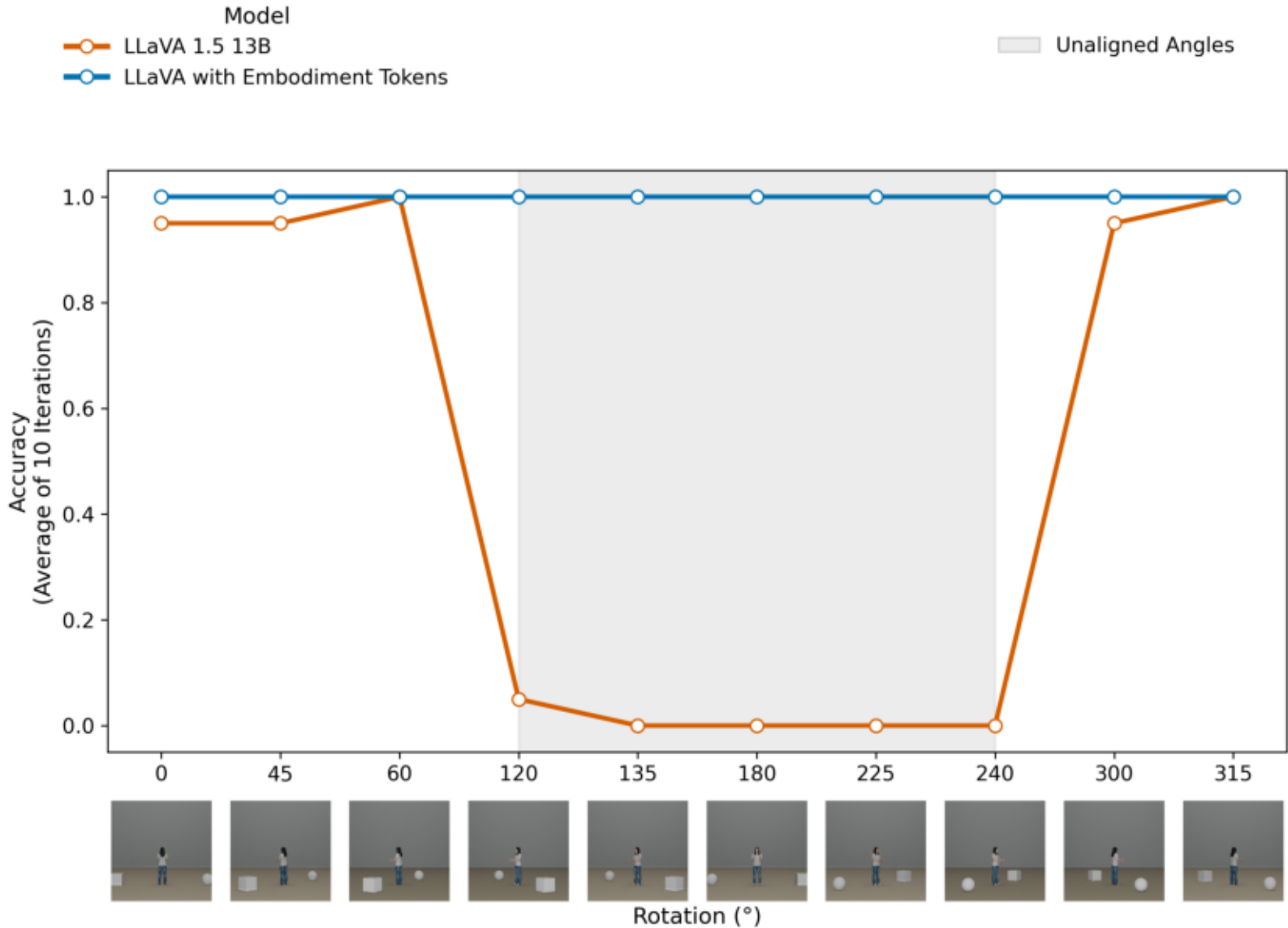
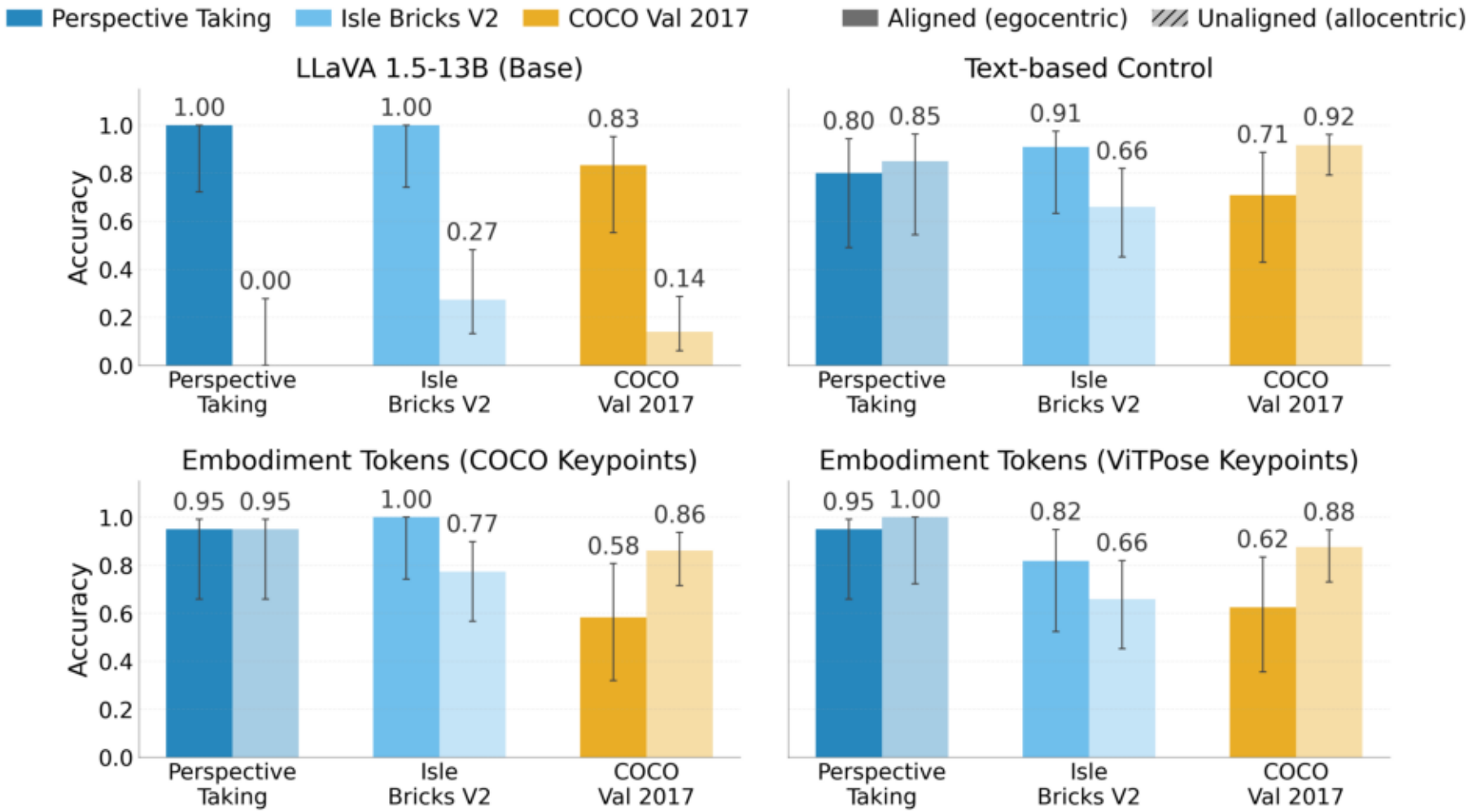


(a)



(b)



(c)

Input (aligned item)

Which side of the person is the cube on from the person's perspective?

LLaVA 1.5 13B (Base)

From the person's perspective, the cube is on their **right** side.

LLaVA with Emb. Text

The cube is right of the person from the viewer's frame. The person's pose is 162.0 173.0 [...] 7 0. Since 7 lies in the 315-360 bin, the person is aligned. After performing a mental rotation of 315-360 degrees, the cube is now to the right. Therefore, the cube is to the **right** from the perspective of the person.

LLaVA with Emb. Tokens

The cube is right of the person from the viewer's frame. The person's pose is <POSE_START> <SHOULDER_L><X_157> <Y_176> [...] <YAW_0> <TORSO_W_1> <ORIENT_END><POSE_END>. Since <YAW_0> lies in the 0-45 bin, the person is aligned. After performing a mental rotation of 0-45 degrees, the cube is now to the right. Therefore, the cube is to the **right** from the perspective of the person.

Input (unaligned item)

Which side of the person is the cube on from the person's perspective?

LLaVA 1.5 13B (Base)

From the person's perspective, the cube is on their **right** side.

LLaVA with Emb. Text

The cube is left of the person from the viewer's frame. The person's pose is 172.0 173.0 [...] 4 1. Since 4 lies in the 180-225 bin, the person is unaligned. After performing a mental rotation of 180-225 degrees, the cube is now to the right. Therefore, the cube is to the **right** from the perspective of the person.

LLaVA with Emb. Tokens

The cube is right of the person from the viewer's frame. The person's pose is <POSE_START> <SHOULDER_L><X_175><Y_171> [...] <YAW_4> <TORSO_W_1><ORIENT_END><POSE_END>. Since YAW_4 lies in the 180-225 bin, the person is unaligned. After performing a mental rotation of 180-225 degrees, the cube is now to the left. Therefore, the cube is to the **left** from the perspective of the person.